Department of Mathematics, IIT Madras
MA 2031 - Linear Algebra for Engineers
End-semester Examination
PART-B

Date: December 14, 2020. Time: 70 minutes Max. marks: 30

Answer all questions. Each question carries 5 marks.

1. Let $V = P_2(\mathbb{R})$ be the vector space of all polynomials of degree not exceeding 2. Let $\mathcal{B} = \{1 - 3t^2, 2 + t - 5t^2, 1 + 2t\}$ and $\mathcal{C} = \{1, t, t^2\}$ be ordered bases for V. Suppose that $T : V \to V$ is the differentiation map.
   (a) Compute $[T]_{\mathcal{C}}$.
   (b) Let A denote the change of coordinates matrix that changes the $\mathcal{C}$ coordinates of an element of V to its $\mathcal{B}$ coordinates. Compute the matrix A.
   (c) Express $[T]_{\mathcal{B}}$ in terms of the matrices A and $[T]_{\mathcal{C}}$.

2. Let $V = P_2(\mathbb{R})$ be the space of all polynomials of degree not exceeding 2, endowed with the inner product $\langle f(t), g(t)\rangle = \int_0^1 f(t)g(t)\,dt$. Suppose that $T : V \to V$ is the linear map defined by $T(a_0 + a_1 t + a_2 t^2) = a_1 t$.
   (a) Demonstrate that T is not self-adjoint by considering $\langle T(1), t\rangle$.
   (b) Let $\mathcal{C} = \{1, t, t^2\}$ and $B = [T]_{\mathcal{C}}$. Show that $B^* = B$.
   (c) Although $B^* = B$, T is not self-adjoint as observed in part (a). Why is this not a contradiction? Justify.

3. Let $V = P(\mathbb{R})$ denote the inner product space of all polynomials with real coefficients, endowed with the inner product $\langle f(t), g(t)\rangle = \int_0^1 f(t)g(t)dt$, for $f, g \in V$. Let $W = P_1$ denote the subspace of polynomials of degree less than or equal to 1. Find the best approximation to $t^2 \in V$ from the subspace W.

4. Let V be an n-dimensional inner product space over $\mathbb{R}$ and $T : V \to V$ be a self-adjoint linear map.

   (a) Prove that eigenvalues corresponding to distinct eigenvalues of T are orthogonal.

   (b) Suppose that $dim(Range(T)) = k < n$. What is the maximum number of distinct eigenvalues T can have? Justify your conclusion.

5. Let $A \in M_3(\mathbb{R})$ denote the matrix all of whose entries are one. Find all the eigenvalues of A and obtain two non-trivial subspaces W of $\mathbb{R}^3$ such that $Ax \in W$ for all $x \in W$.

6. Let V be a finite dimensional vector space over $\mathbb{C}$ and $T : V \to V$ be a linear map.

   (a) State the Cayley-Hamilton theorem for the linear map T.

   (b) The linear map $T : V \to V$ is said to be nilpotent if $T^k = 0$ for some $k \in \mathbb{N}$. Using part (a) or otherwise, prove that $T : V \to V$ is nilpotent if and only if the only eigenvalue of T is 0.

***